

INTERNSHIP FINAL PRESENTATION: SERENA LIN

Controls & Automation Intern @ Tesla
May - August 2026



Introductions

- B.S. Electrical and Computer Engineering at Northeastern University, Boston MA
- Candidate for M.S. Electrical Engineering, Power Systems & Power Electronics
- Prior Experience
 - Bose Corporation, Electrical Engineering Co-op, Framingham, MA
 - Johnson & Johnson, Surgical Robotics Electrical Engineering Co-op, Santa Clara, CA
 - NSF-REU at UMBC Mobile Pervasive Sensing and Computing Lab, Baltimore, MD
 - Experiential Robotics Lab at Northeastern University



Projects & Contributions Overview

- Controls development on manufacturing lines for component molding
 - Programming: Gantry pick & place
 - Debugging: Electrical, Software & Mechanical
 - Hardware Commissioning
 - Testing & Validation
- Laser Light Tower Safety Device
 - Electrical schematic design
 - Panel Build
 - Safety Zone Configuration
 - Test & Deploy

COMPONENT MOLDING AUTOMATION DESIGN

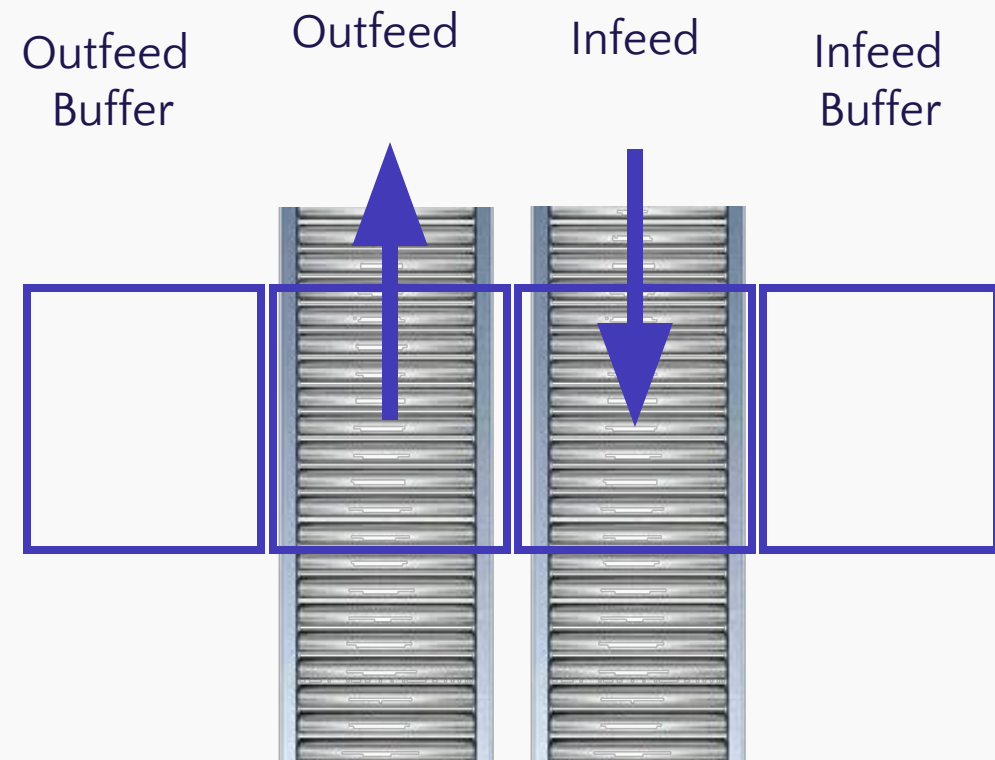
Programming
Debugging
Hardware Commissioning
Testing & Validation

Machine Overview

- Pallet based conveyance
- Heating mechanism
- Plastic molding parts
- Precision loading and unloading

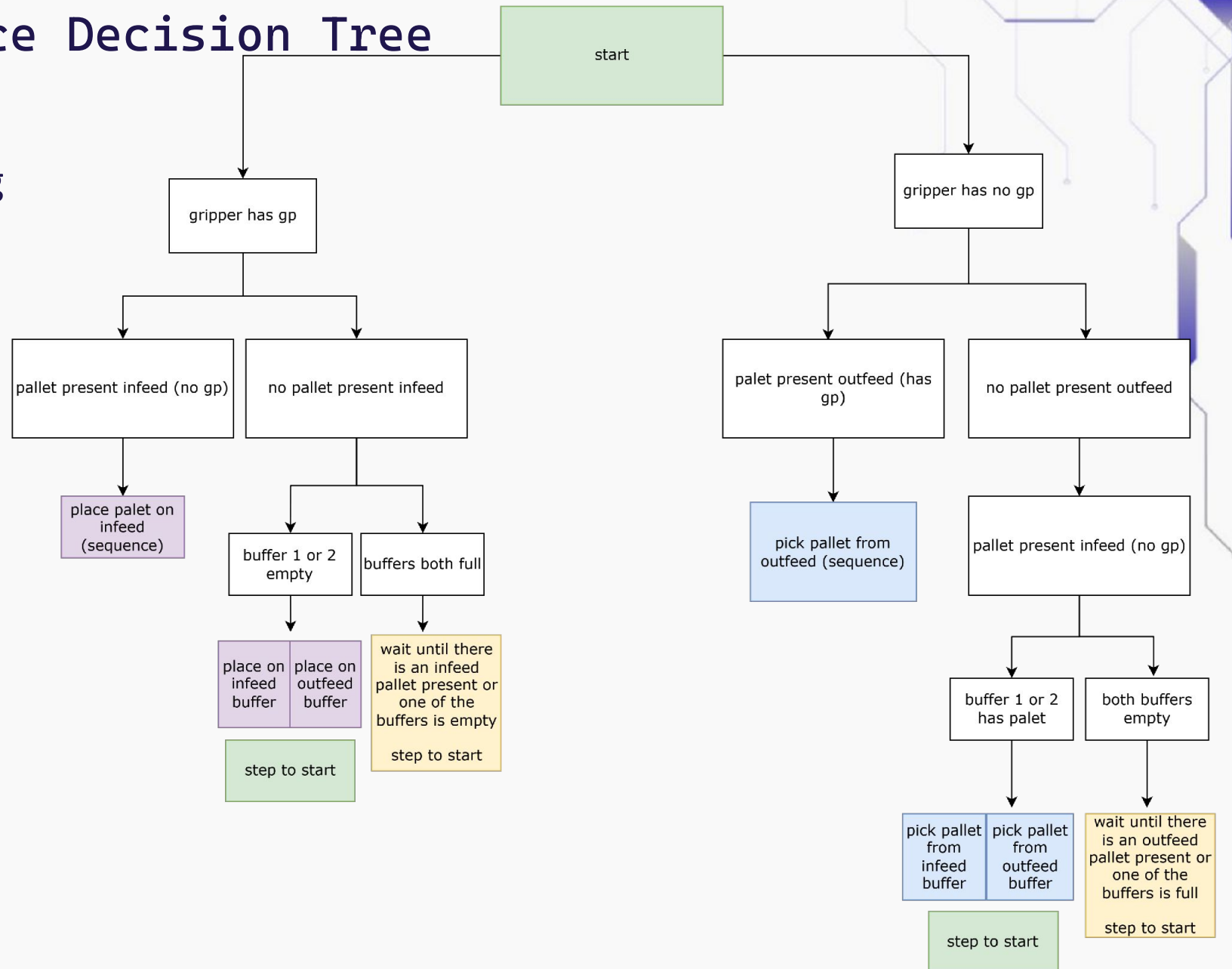
Precision Unloading: X-Z Gantry Pick and Place

Problem Statement: Remove outfeed pallet guide plate and place upon infeed pallet, using buffers as necessary

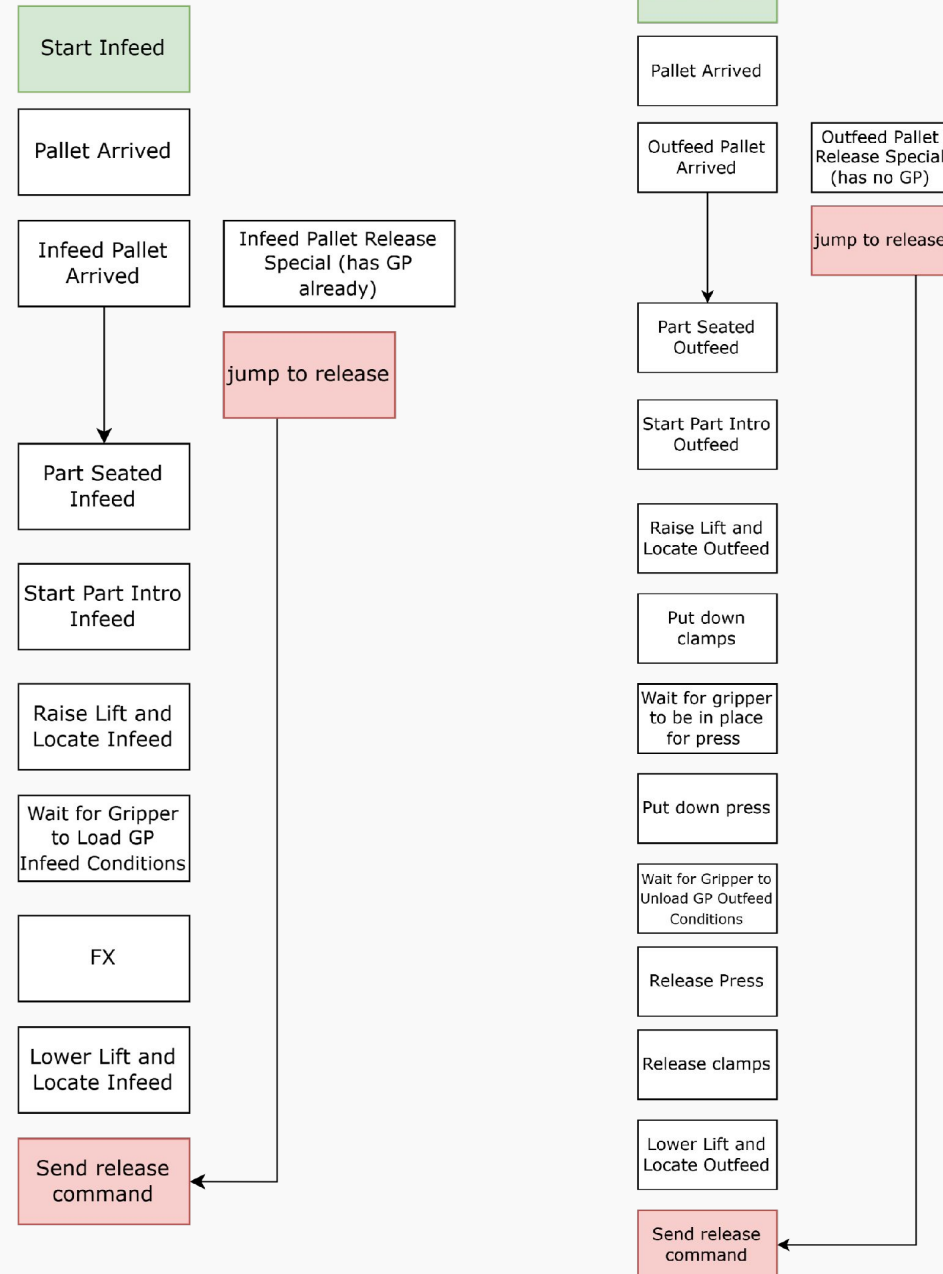


Gantry Pick and Place Decision Tree

- Race conditions from multiple sequences starting at the same time
 - Gantry crashes
- Adding interlocks and supervisions for safety
- Edge cases
- Teaching positions
- Mechanical adjustments



Gantry Pick and Place Infeed & Outfeed



Gantry Dry Cycle



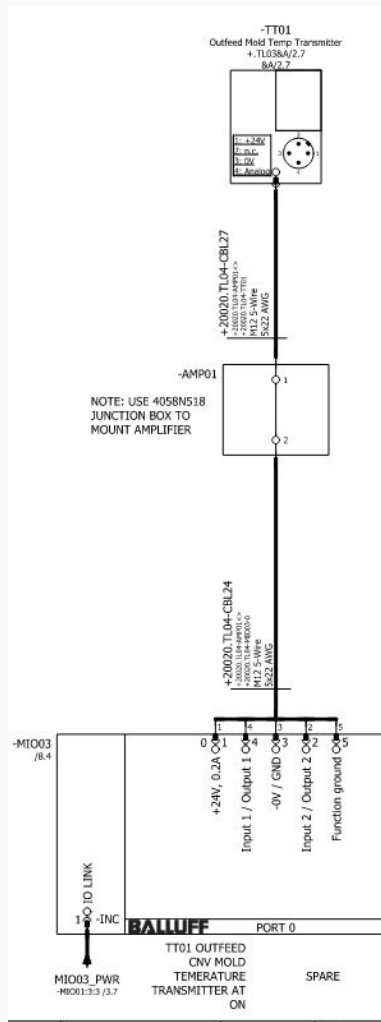
Debugging: 400+ I/O Points

- Analog, digital, & IO link sensors
- Input Checks: Sensor Inputs
 - Positional and proximity
 - Photo eye lasers
 - Analog temperature
- Output Checks: Triggering Actuators
 - Valve banks
 - Lift and locate
 - Lift and transfer units
 - Oven door
 - Gantry grippers
 - Hold down tooling

Common Issues Encountered

- Work and home proximity sensors wiring switched on Balluff block
- Balluff MIO block input ports misconfigured for device type
- Work and home output actuator air tubes flipped
- HMI tags mismatched
- Hardware mismatch from PLC

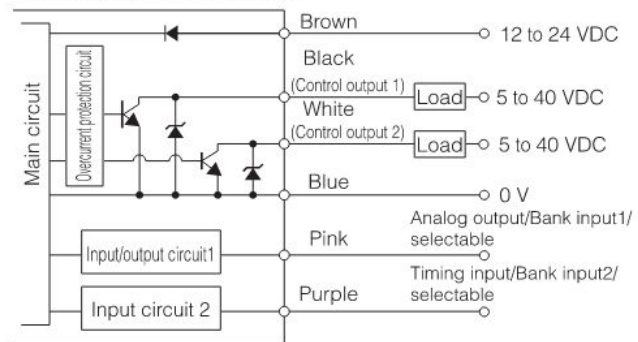
Electrical Debug: Analog Temperature Sensor



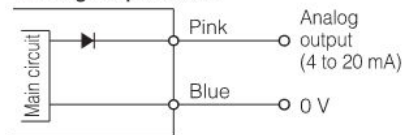
Input/Output Circuit Schematic

■ FT-50AW/55AW

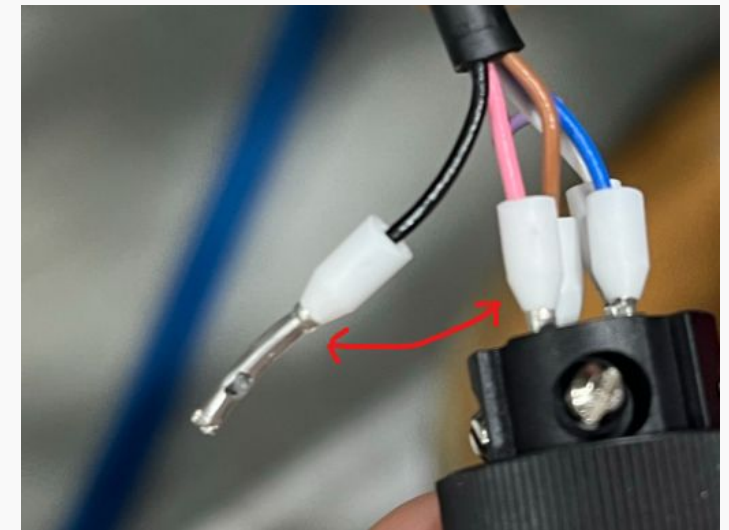
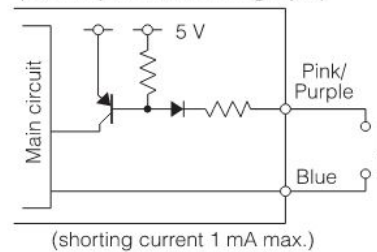
Input/output circuit schematic



Analog output circuit



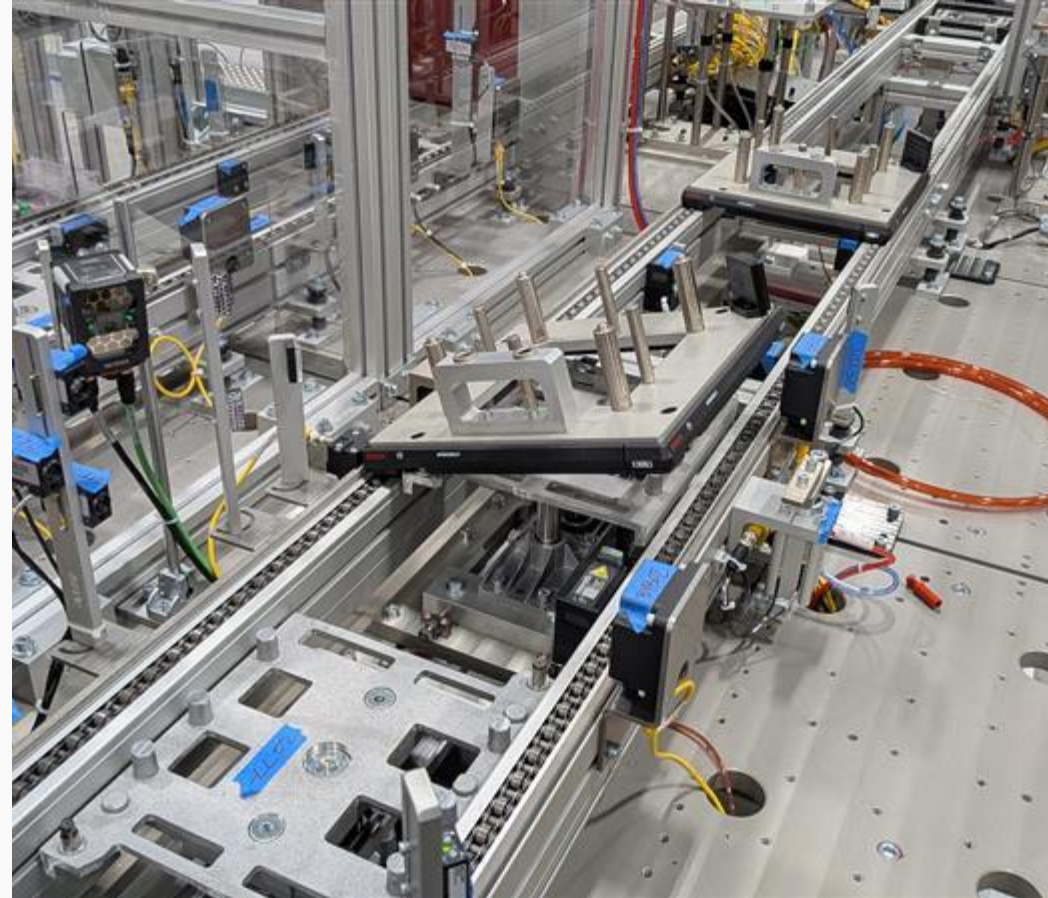
External input circuit (Bank input 1 & 2, Timing input)



From FT-50AW Manual

Mechanical Debug

- Prox and photo eye Alignment
- LTU alignment with conveyor
- Pallet stop alignment with lift
- Height and speed of lift



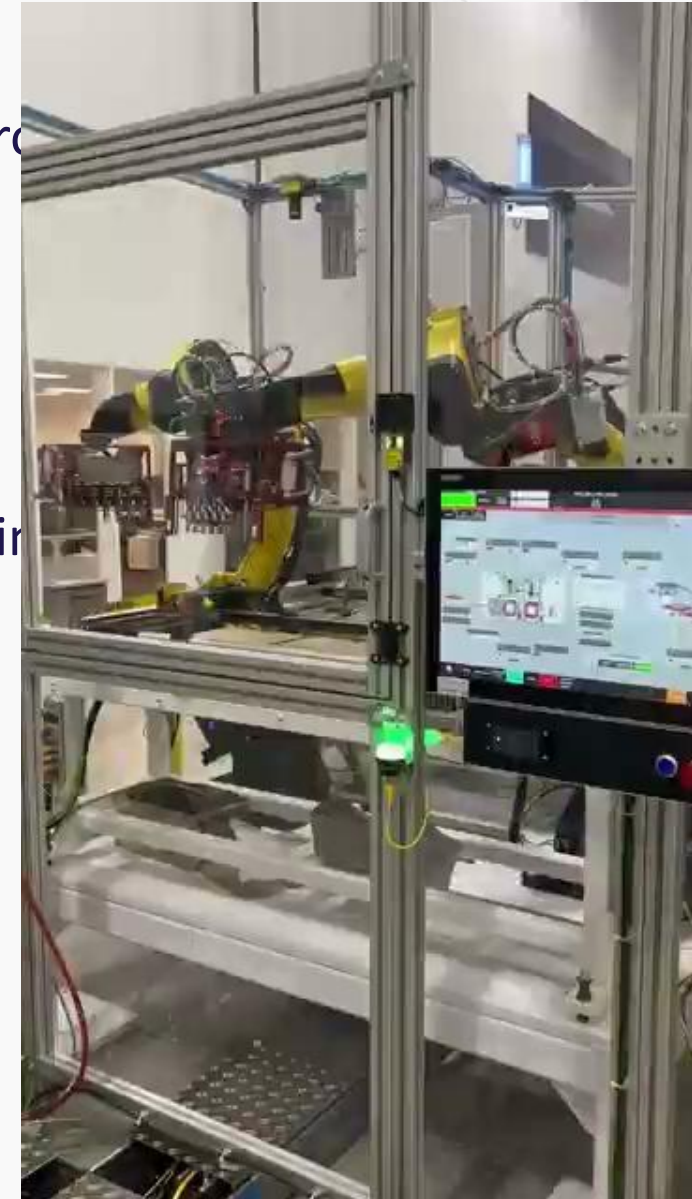
Debugging: Pallet Cycle

- Creating a new HMI mode for Pallet Cycle
- Checking mechanical issues and conveyor function
- Investigate pallet stop logic
 - Faulted, ready to receive, ready to release
 - Timer timeout and stored memory
- Learned about part intro and structure of stations

Request to Enter Door, Button, & Lock

Purpose: entering the cell to remove bad parts, request entry from

- Add a function block for the light and push button
- Test I/O signals and safety signals
 - Button Pressed
 - Locking
- Electrical debug of safety signals: additional Murr block was in
 - Door closed safety signals



Hardware Commissioning

Common Tasks

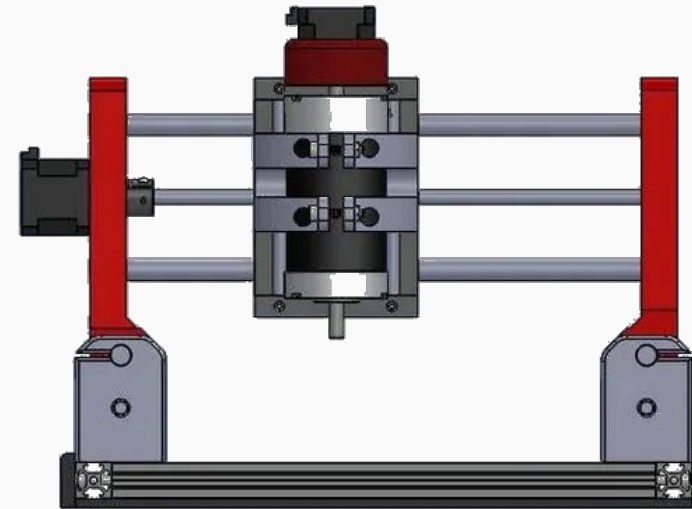
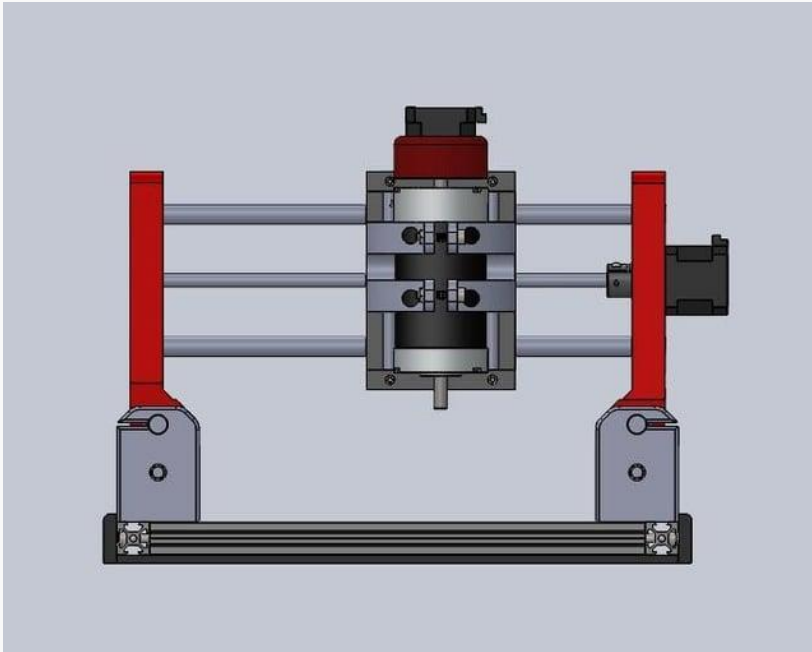
- Assigning profinet names & IP addresses
- Setting dip switches, port settings, and F-Address parameters
- Updating firmware

Types of Hardware

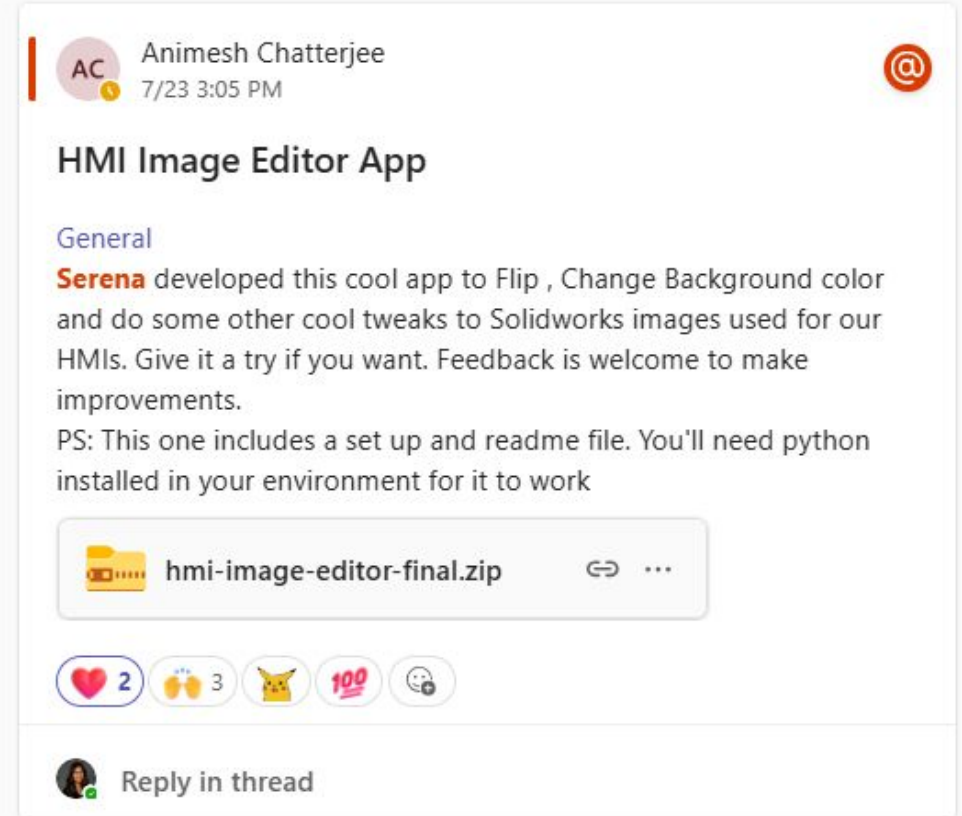
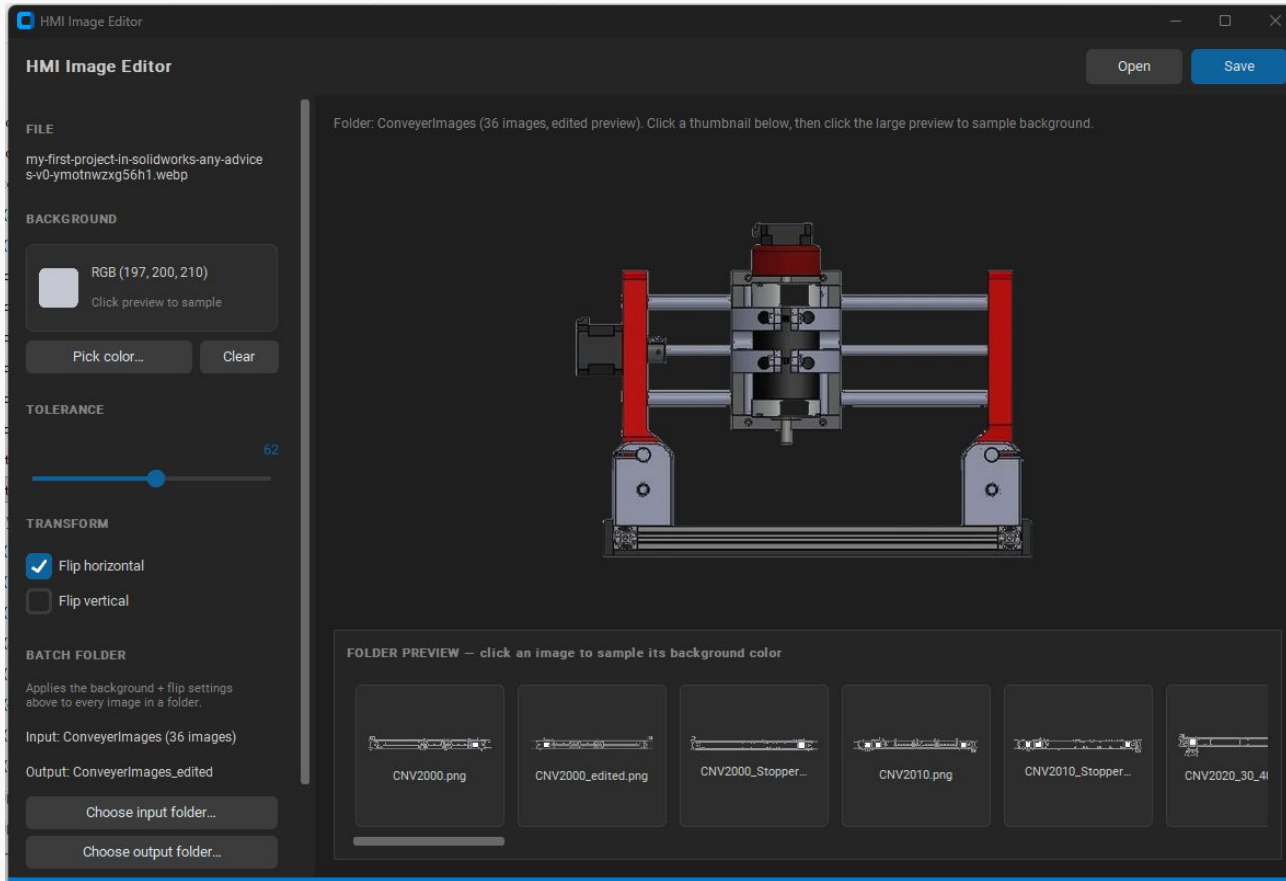
- Murr Safety Blocks
- Balluff MIO Blocks
 - Port 2 & 4, I-O Link & Digital Input Output
- HMI Screens
- Euchner Safety Gates

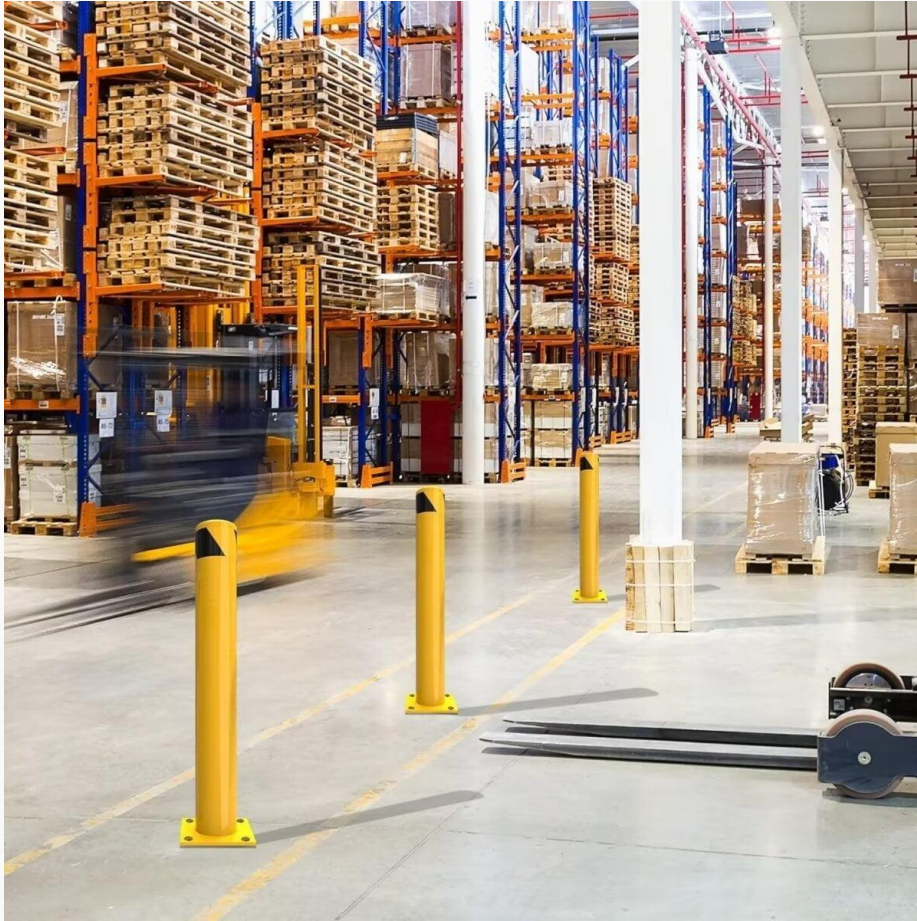
HMI Image Processing

- FAS1 & FAS2 Mirrored Lines



Development of HMI Image Processing Tool: Cursor





LASER LIGHT TOWER SAFETY DEVICE

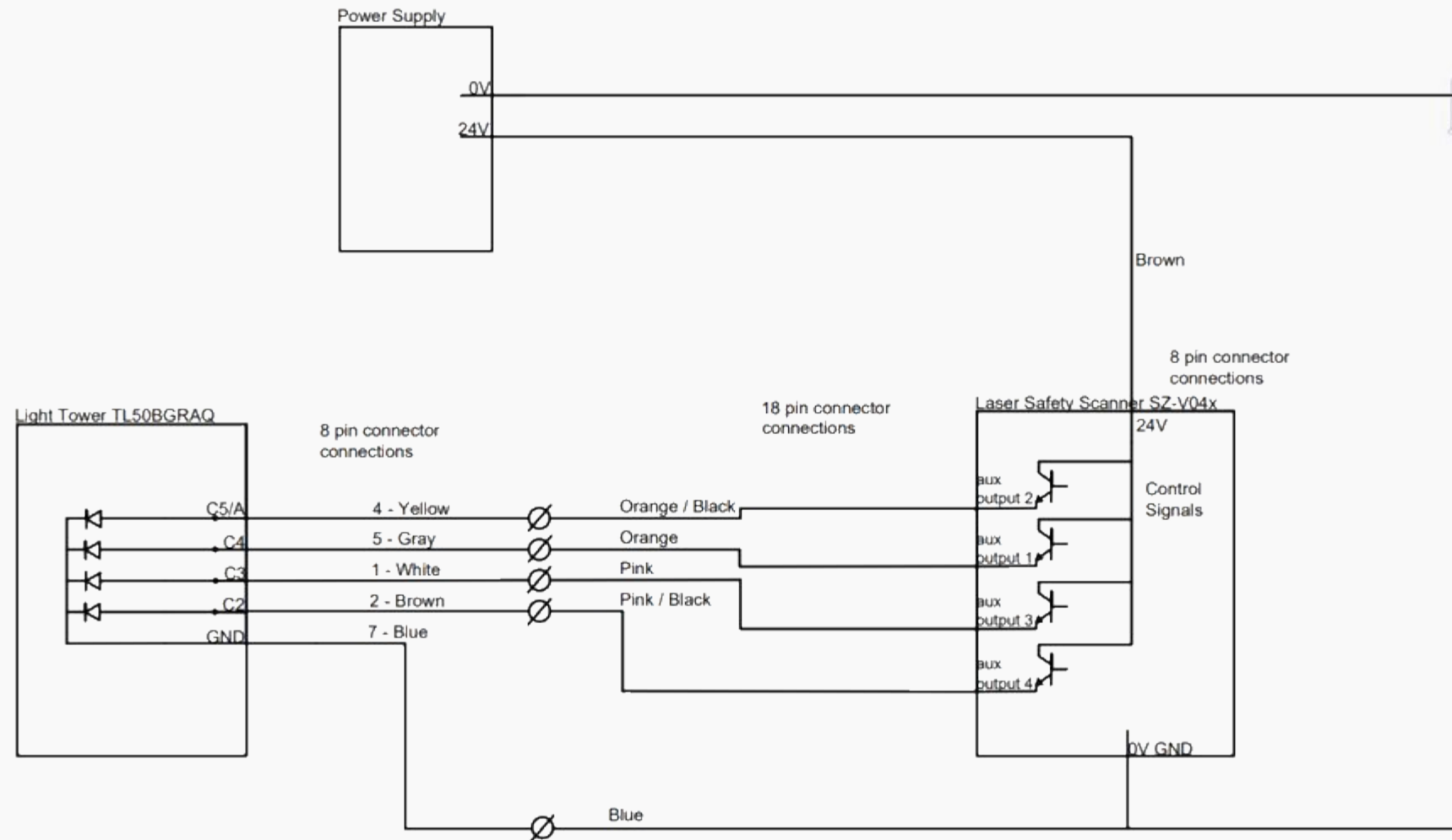
Electrical Schematic Design
Panel Building
Safety Zone Configuration
Testing & Deployment

Problem Statement

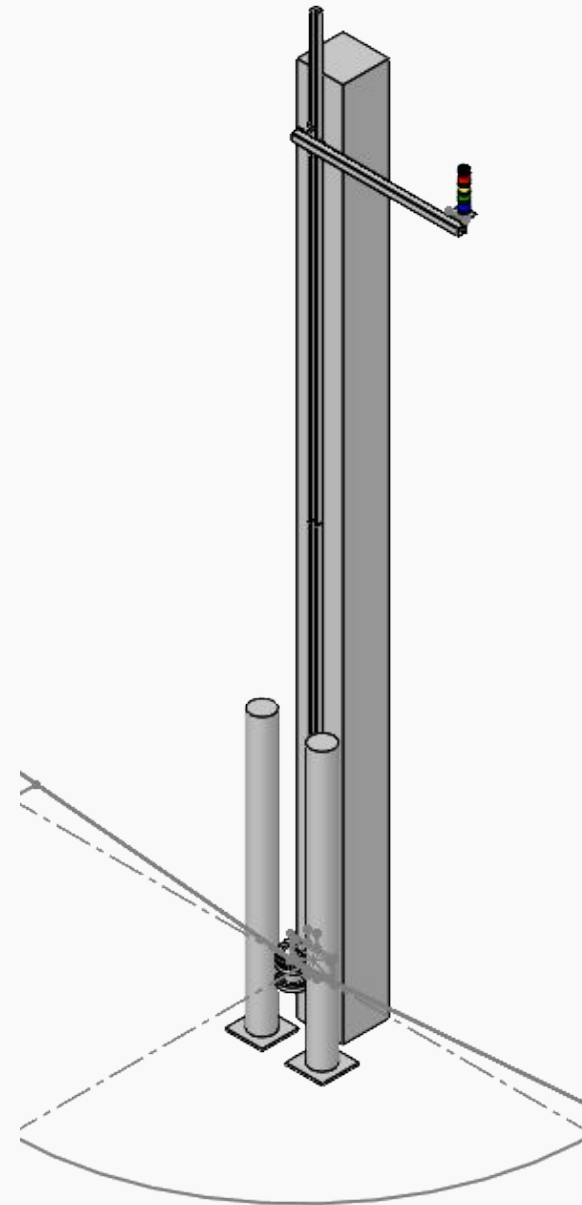
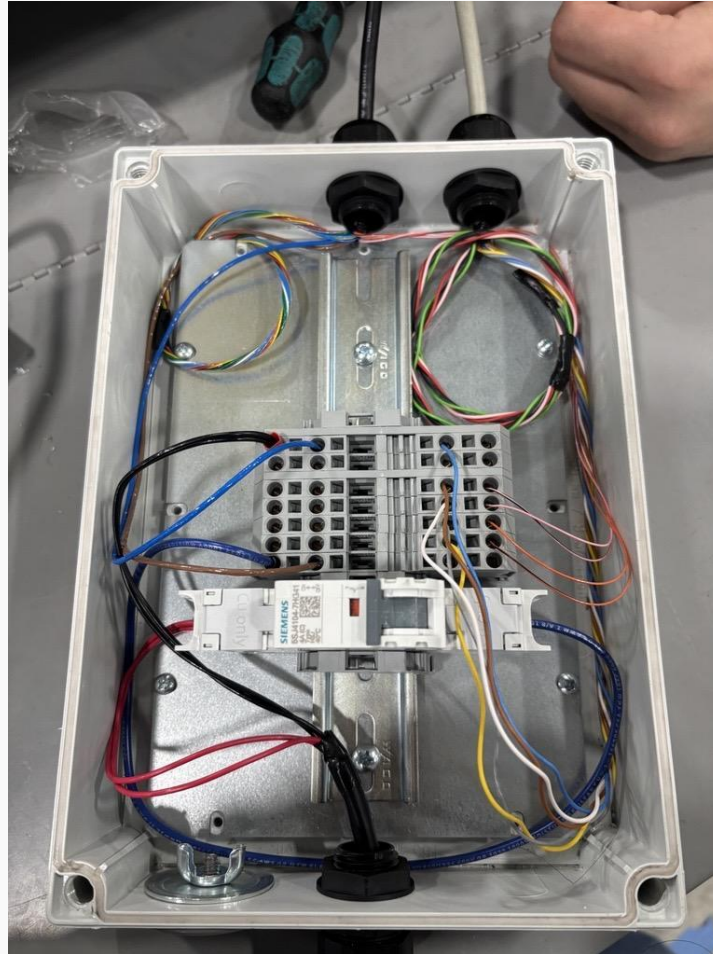
- Protect load bearing pillar from delivery trucks
 - Interface a SV-4 Safety Laser Scanner with a Banner Tower Light & Alarm



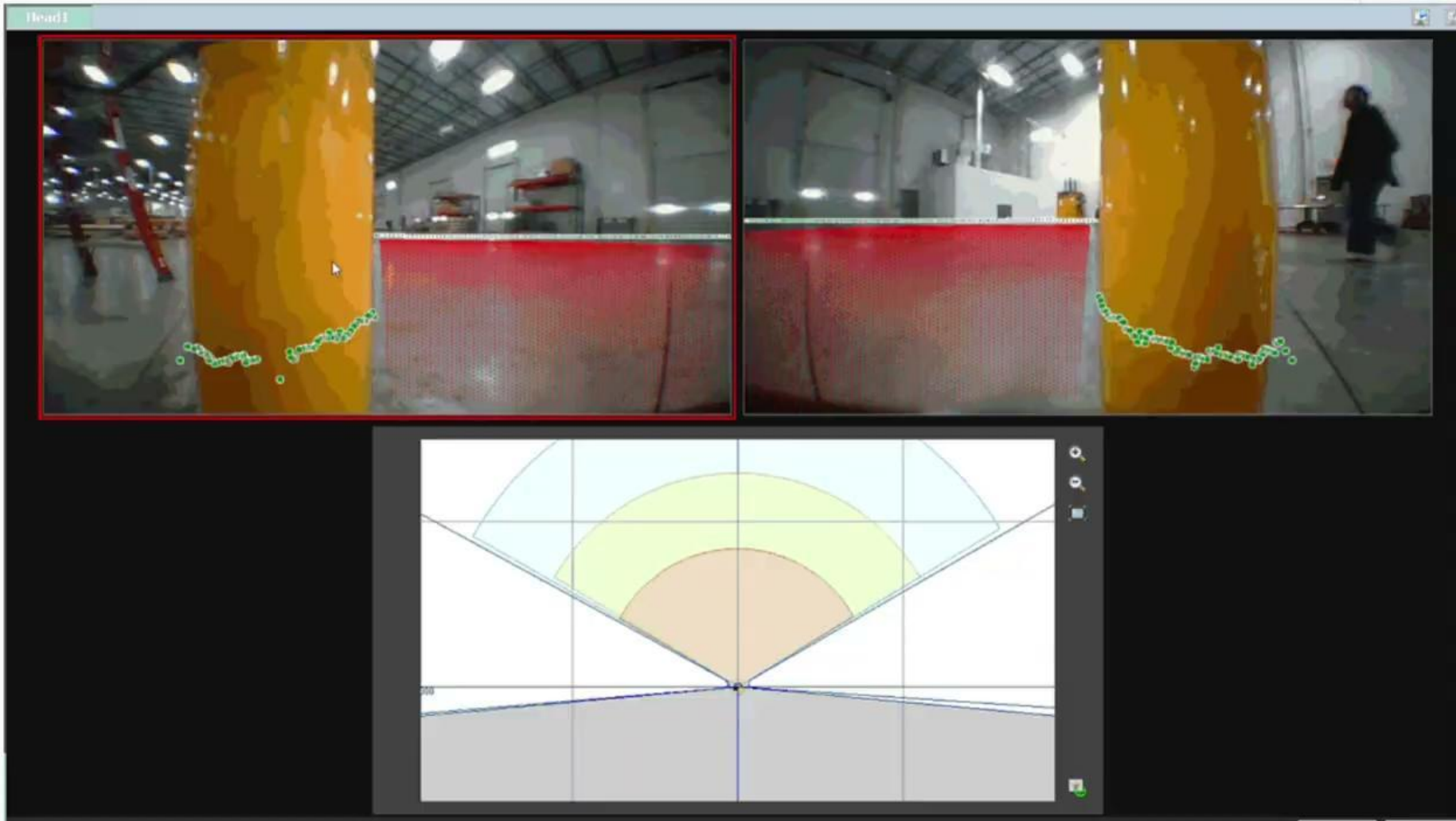
Electrical Schematic Design



Panel Build & Mechanical Mounting



Safety Zone Configuration



Input & Output Configuration

Setting

Monitoring

History

Unit Config.

Operation

Zone

Others

Transfer

State information output

State information output Not used

Auxiliary outputs

AUX1 Detection in prot. zone

AUX2 Detection in prot. zone

AUX3 Detection in warn. zone (A)

AUX4 Detection in warn. zone (B)

AUX5 Not used

AUX6 Not used

Wire color and assigned function

Display settings

Power saving mode Not used

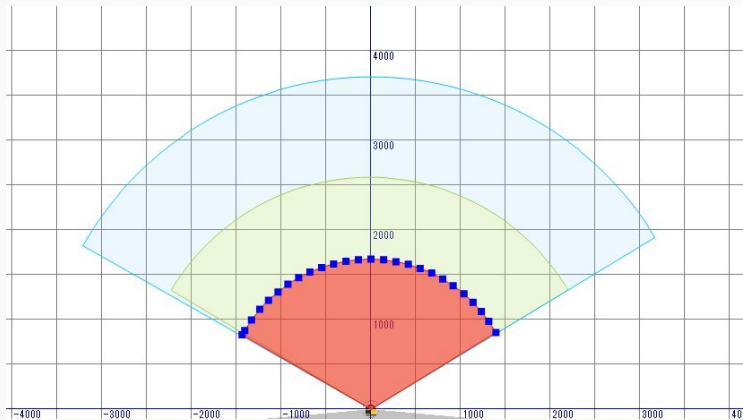
Camera blur Not used

Detection history

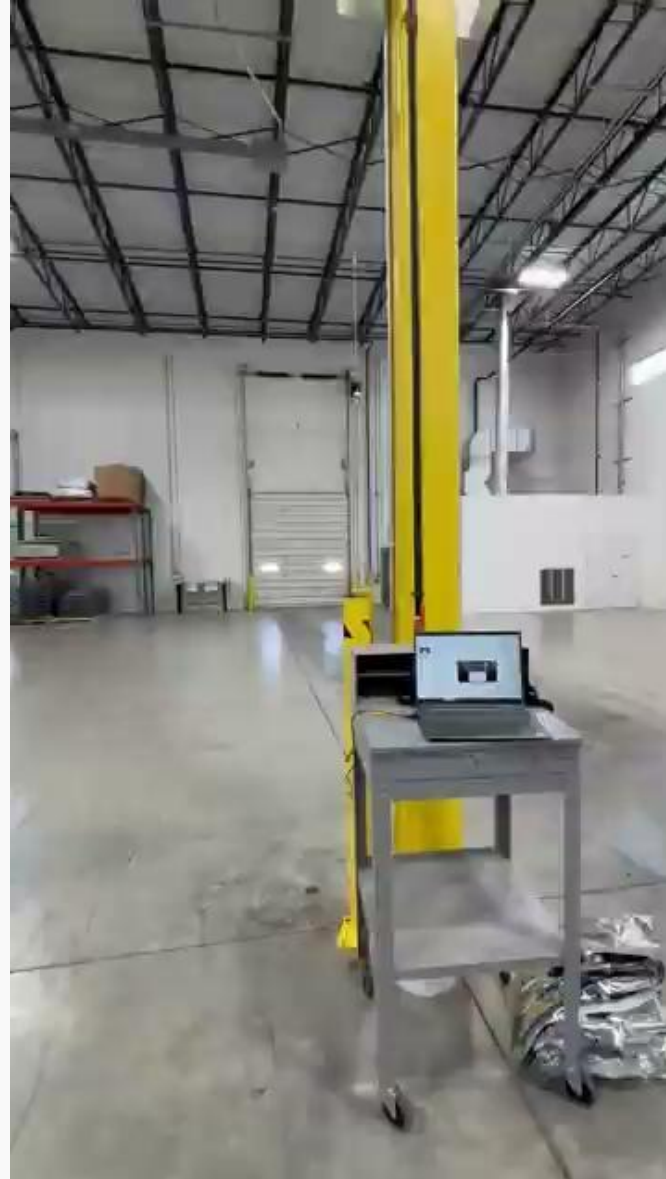
Saving mode Single Frame/Photo

Wire color and assigned function

Wire color	Assigned function
Brown	+24V DC
Blue	0V
Black	OSSD1
White	OSSD2
Gray	Not used
Gray/Black	Not used
Yellow	Input 1: Reset input (1/2) or Laser OFF input
Red	Input 2: EDM input(1/2)
Light blue	Input 3: Not used
Light blue/Black	Input 4: Not used
Yellow/Black	Input 5: Not used
Red/Black	Input 6: Not used
Orange	AUX output 1: Detection in prot. zone
Orange/Black	AUX output 2: Detection in prot. zone
Pink	AUX output 3: Detection in warn. zone (A)
Pink/Black	AUX output 4: Detection in warn. zone (B)
Green	AUX output 5: Not used
Green/Black	AUX output 6: Not used



Testing & Deployment



LEARNINGS

Challenges & Learning

Challenges

- First experience with PLCs and automation
- Not knowing what you don't know
- Understanding the layers of connections
 - Electrical schematics
 - Device hardware config
 - Profinet communication and networking
 - Safety signals
 - PLC tags and digital input outputs
 - Pneumatic actuation and air tubing
 - Graph sequences and code structure
 - HMI screens

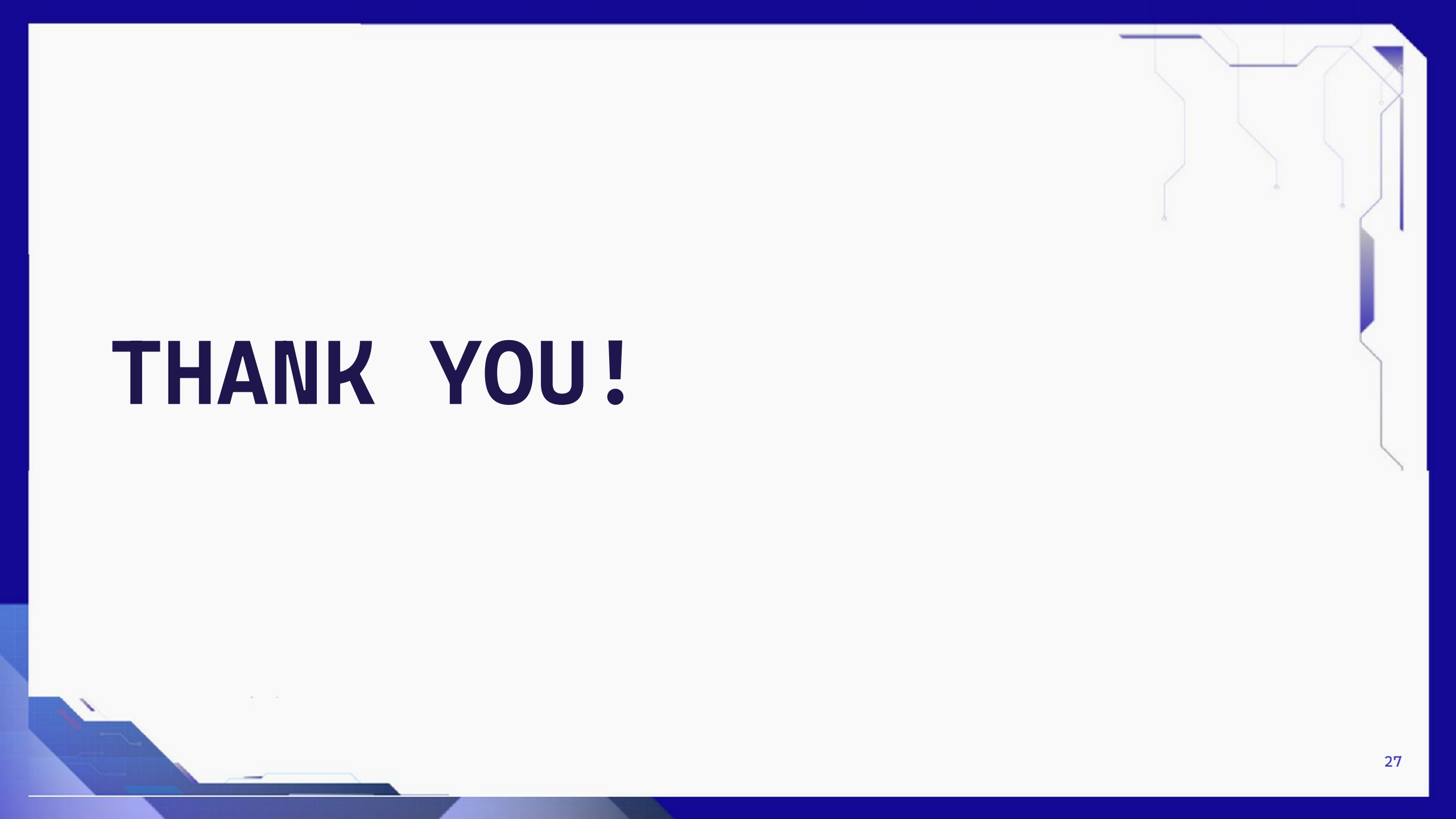
Technical Skills

- Siemens ladder logic
- IP addressing and Profinet
- Air Actuation
- Tag tables and connections to schematics
- TIA graph sequences & interlocks
- HMI screens & faceplates
- Hardware Debug
- Second line I/O checks in 3 days vs 3 weeks for the first line

General Takeaways

- Designing with flexibility for future innovation
- Writing maintainable code
- Jumping into things and making mistakes is the best way to learn
- Confidence in fixing things & finding issues





THANK YOU!